FULL NAME: ______________________________ **BLOCK:** ____________

Inverse Variation Applications + DOK-3 Performance Task

Lesson 4-1 · Period 2 · Teacher Edition

OBJECTIVES

Math Objective	Apply inverse variation to real-world contexts (wavelength, gas pressure, travel distance), and distinguish DIRECT variation ($y = kx$) from INVERSE variation ($xy = k$) inside a single multi-part problem.
Language Objective	Students will use the frame "This is ___ variation because as ___ ___, ___ ___ proportionally." to justify whether a relationship is direct or inverse.
Essential Understanding	Not every covariation is inverse variation. Direct variation has a constant RATIO ($y/x = k$); inverse variation has a constant PRODUCT ($xy = k$). A single scenario can contain BOTH types across different variable pairs.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students model 4 contextual inverse-variation problems, then tackle a Performance Task (Ramón's road trip) that embeds direct and inverse variation in one scenario. Every item is Savvas-sourced.
Essential Question	How does inverse variation model real-world relationships, and how is it different from direct variation?
Materials	Pencils; student packet; calculator. No Chromebooks required.

[PIN] PRE-FLIGHT — P2 IS THE DOK-3 SPINE DAY

Today's conceptual core is Practice #26 (Ramón). Part A is direct variation, Part B is inverse — same scenario, different variable pairs. That contrast IS the lesson's deepest content.
Pace: Try-It 3 (5 min) → #12 (4 min) → #17 (5 min) → #23 (7 min) → #26 (12–15 min). ∼33–36 min total Explore.
If running tight: cut #12 (the HOT inverse-square is tangent, not central). Never cut #26.

Tag: bridge from P1

Do Now **DOK:** 1 **Minutes:** 5

Teacher does

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student for xy computation.

Students do

- Compute xy for each column.
- State k if constant; explain if not.

Questions to ask

- What do you multiply to check inverse variation?
- Is xy constant here? What does that tell you?

Adult role
Solo teacher: circulate silently. No hints.

Do Now (4-1-savvas-q15)

15. Do the tables of values represent inverse variations? Explain. See Example 1. Table - x: $1, 2, 3, 4, 5, 6$; y: $60, 30, 20, 15, 12, 10$.

x	1	2	3	4	5	6
y	60	30	20	15	12	10

[CHECK] ANSWER KEY

Answer key (from TE): Yes; as the x-values increase, the y-values decrease. Also the product of each pair of x- and y-values is 60 ($k = 60$, so $y = 60/x$).

Tag: teacher models Example 3

Launch **DOK:** 2 **Minutes:** 12

Teacher does

- Project Example 3 (bouzouki). Think aloud: extract the two quantities (s = string length, f = frequency). Compute $k = sf$.
- "Now I can find f for any s. Plug in the new length, solve."
- Flag the Application Rules card — print it in student minds.
- 30-sec pause: "What's k for YOUR Do Now table?" (partner check)
- Do NOT solve Try-It 3 or Practice items — release at minute 17.

Students do

- Watch the bouzouki example silently.
- During pause: state k from Do Now to partner.
- Copy Application Rules card into margin.
- Note the DIRECT vs INVERSE warning: this comes back in #26.

Questions to ask

- What two quantities are related? What's constant?
- "Every radio wave with frequency f has wavelength w such that wf = constant." Why is that inverse?
- If frequency doubles, what happens to wavelength? Why?

Adult role
Project. Think aloud. Release promptly at minute 17.

[MIC] TEACHER SCRIPT

1. "Watch me set up a real-world inverse variation. Bouzouki string: length s varies inversely with frequency f."
2. "Step 1: Identify the quantities and find k. $k = sf = 26 \times 329.63 = 8{,}570.38$."
3. "Step 2: Write the equation. $s = k/f$, so $s = 8570.38/f$."
4. "Step 3: Plug in the new value. For $s = 13$: $f = 8570.38/13 = 659.26$ cycles/sec."
5. "Tonight's Performance Task has a twist — ONE scenario contains BOTH direct and inverse variation. Be ready to tell them apart."
6. Release to Explore.

Tag: TPS + DOK-3 driver

Explore **DOK:** 2–3 **Minutes:** 33

Teacher does

- 5 laps across 33 min.
- Lap 1 (5 min): Try-It 3 — ice cube. Press pairs to identify k = Temp $\times$ Time.
- Lap 2 (4 min): #12 — inverse square. This is a THINKING item; let pairs struggle.
- Lap 3 (5 min): #17 — graph reading + context. Confirm $k = 300{,}000$.
- Lap 4 (7 min): #23 — Boyle's Law. Check that pairs compute volleyball's $k = 1350$.
- Lap 5 (12 min): [STAR] #26 Ramón. PART A is direct; PART B is inverse. Press hard on that contrast.
- Do NOT give #26 answers. Pairs present Part A/B to each other; one pair presents at Share.

Students do

- 5 items in order: Try-It 3 → #12 → #17 → #23 → #26.
- For #26: BEFORE solving, identify which part is direct and which is inverse. Write it down.
- Justify with sentence frame.

Questions to ask

- What's k in this problem? What units?
- #12: $y = k/x^2$. If $x \to 4x$, what happens to y? (Divide by 16.)
- #17: from the graph, what's k? How do you read it?
- #23: Whose k do we use — the volleyball's, or the basketball's? (Volleyball's; the basketball only gives the target PRESSURE.)
- #26 Part A: is distance/time constant? → direct variation.
- #26 Part B: is gas $\times$ time constant? → inverse variation.

Adult role

Solo teacher: 5 laps. Press DIRECT vs INVERSE distinction on #26.

Try It 3 (4-1-savvas-try-it-3-lesson-4-1-matches-examp)
Try It 3. The amount of time it takes for an ice cube to melt varies inversely with the air temperature in degrees Celsius. At 20°C, the ice will melt in 20 minutes. How long will it take the ice to melt if the temperature is 30°C?

[CHECK] ANSWER / ANTICIPATED TALK

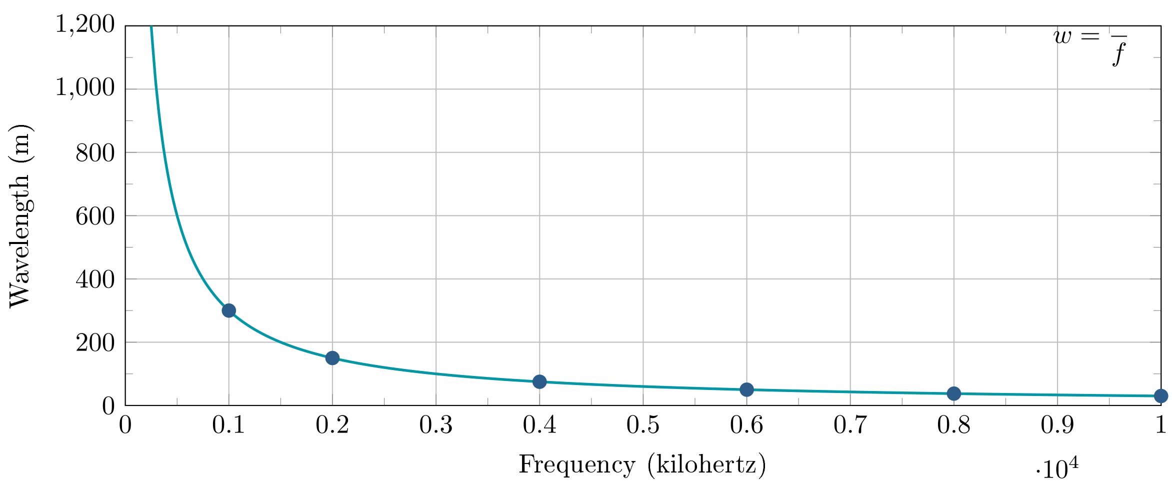
Answer key (from TE): ≈ 13.3 minutes. Reasoning: $k = 20 \cdot 20 = 400$; $t = 400/30 \approx 13.3$.

Practice #12 (4-1-savvas-q12)
12. Higher Order Thinking: Suppose y varies inversely as the square of x. If x is multiplied by 4, what is true for the value of y?

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): Divide by 16. Reasoning: $y = k/x^2$. When $x \to 4x$, new $y = k/(4x)^2 = k/(16x^2) = (1/16)y_{\text{original}}$.

Practice #17 (4-1-savvas-q17)
17. The wavelength, w, of a radio wave varies inversely to its frequency, f, as shown in the graph. The graph shows w (m) on the y-axis from 0 to 1200, and f (kilohertz) on the x-axis from 0 to 10000, with the curve $w = k/f$ passing through data points including (1000, 300), (2000, 150), (4000, 75), (6000, 50), (8000, 37.5), (10000, 30). A radio wave with a frequency of 1,000 kilohertz has a length of 300 m. What is the frequency when the wavelength is 375 m?



[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): 800 kHz. Transcription and graph description verified against Gemini 3.1 Pro (LaTeX output with $w = 300000/f$ curve and data points).

Practice #23 (4-1-savvas-q23)
23. Model With Mathematics: Boyle's Law states that the pressure exerted by fixed quantity of a gas, P, varies inversely with the volume the gas occupies, V, assuming constant temperature. The volume and air pressure of a volleyball are 300 in^3 and 4.5 psi. The volume and air pressure of a basketball

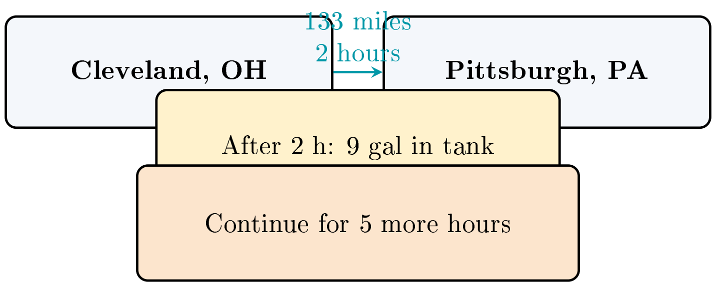
are 415 in^3 and 8 psi. How much smaller would the volleyball have to be to be equal the air pressure of the basketball?

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): 131.25 in^3. Volleyball $k_V = 300 \times 4.5 = 1350$. At $P = 8$ psi, $V = 1350/8 = 168.75$ in^3. Difference $= 300 - 168.75 = 131.25$ in^3. (Note: the basketball's own k is irrelevant to the answer - we only use basketball's $P = 8$ psi as the target for the volleyball.)

Practice #26 [STAR] DOK-3 DRIVER (4-1-savvas-q26)

26. Performance Task. Suppose Ramón takes a road trip. He starts from his home in the suburbs of Cleveland, OH and travels to Pittsburgh, PA to visit his aunt and uncle.
Part A: The distance Ramón drives from Cleveland to Pittsburgh is 133 miles. The trip takes him 2 hours. The distance d in miles that Ramón drives varies directly with the amount of time t in hours he spends driving. Write the equation of the direct variation. Use the given relationship and the equation to find the number of miles Ramón would travel if he continues on for 5 more hours.
Part B: The amount of gas in Ramón's car is 9 gal after he drives for 2 h. The amount of gas g in gallons in his tank varies inversely with the amount of time t in hours he spends driving. Write the equation of the inverse variation. Use the given relationship and the equation to find the number of gallons in Ramón's tank after 5 more hours of driving.



[CHECK] ANSWER / ANTICIPATED TALK

Transcription verified against Gemini 3.1 Pro (clarified "5 more hours" vs my initial "3.5" misread). **Answer key (from TE):** Part A $d = 66.5t$; 465.5 mi (total $t = 7$ hours since $2+5$). Part B $g = 18/t$; approximately 2.57 gal (at $t = 7$). STRONG DOK-3 driver candidate - the direct-vs-inverse contrast is pedagogically richer than the pure-inverse items #20/#22/#25. Recommend this as the lesson's single-DOK-3 spine.

Tag: academic conversation + Wed bridge

Share / Summary **DOK:** 2 **Minutes:** 5

Teacher does

- Cold-call 1 pair to present Ramón's Part A (direct) + Part B (inverse) using sentence frame.
- Class signals thumbs. Write the direct-vs-inverse test on board.
- Wed bridge: "Tomorrow we graph these inverse functions and name the features."

Students do

- Presenting pair uses sentence frame for Parts A and B.

- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: how do you test direct vs inverse?
- Wednesday preview: what shape do you think $y = k/x$ graphs as?

Adult role
Cold-call a pair that struggled on #26 — hold them accountable to the direct-vs-inverse distinction.

Tag: summary recap (not graded)

Exit Ticket **DOK:** 1–2 **Minutes:** 3

Teacher does

- Collect at door.
- Scan #3 (how to distinguish direct from inverse). Plan P3 Do Now if many students are still fuzzy.

Students do

- 3 sentence stems, honest.

Questions to ask

- Did students articulate the test (y/x constant = direct; xy constant = inverse)?

Adult role
Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

Today I learned that ___.
The biggest trap in Ramón's task was ___.
I distinguish direct from inverse variation by ___.

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min for P2. Use the extra 10 min on Explore #26 (Ramón) — the richer discussion, the better the DOK-3 payoff.
If finished early: hand out Practice #25 (SAT/ACT MC) as fast-finisher.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Application Rules card as a sticker/cut-out.
- For #26, provide pre-formatted work space: "PART A: distance = ___ × time" etc.
- Accept verbal sentence-frame completion in lieu of written.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: "frequency" = how often; "volume" = space inside; "pressure" = push.
- "Performance Task" = an assessment task with multiple parts.
- Pair ELL students with English-dominant partners for TPS.